Finite-Certificate Kernel Transfer in Clique Filtrations: Exact Multiplicity and True Normalized Persistence

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Abstract

For an inclusion of clique complexes $X_1 \subseteq X_2$, true normalized persistence is the fraction

$$\frac{\operatorname{rank}[H_d(X_1) \to H_d(X_2)]}{\dim H_d(X_1)}.$$

Hardness reductions for this quantity must control three features simultaneously: exact homology multiplicities, the inclusion-induced map, and an inverse-polynomial gap above the complete kernel of each endpoint Laplacian. We give a finite-certificate transfer theorem that supplies all three for a fixed palette of weighted clique gadgets. Its main estimate applies to every geometric chain and bounds squared distance to an embedded degenerate logical kernel by

$$C\left(t\lambda^2 + \frac{\langle x, \Delta x\rangle}{g\lambda^{26}}\right).$$

The gadget weight λ is independent of the logical gap g, yielding a geometric gap linear in g. Exact filling identifies the endpoint homology as a quotient V/W_A; nested term sets induce the natural quotient epimorphisms, so persistent rank equals the later logical-kernel dimension. Applying the theorem to an exact real-gate perfect-completeness verifier gives weighted instances with $\beta_d(X_1) = 8$, persistent rank 6 in YES instances and 1 in NO instances, and inverse-polynomial whole-kernel gaps. Thus additive true normalized persistence is hard for the gate-dependent class $\mathsf{BQP}_1^{G_2}$, where $G_2 = \{X, \mathsf{CX}, \mathsf{CCX}, H \otimes H\}$. A common-copy construction transfers the result to unweighted clique filtrations. The theorem does not imply ordinary BQP, unrestricted SDQC_1, or gate-independent BQP_1 hardness.

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1 Introduction

Persistent homology asks which topological features survive as a simplicial complex grows [12]. For a two-step filtration $X_1 \subseteq X_2$, the basic numerical invariant is the rank of the induced map $H_d(X_1) \to H_d(X_2)$. Dividing this rank by $\beta_d(X_1)$ gives an interpretable fraction: among the degree-d classes present initially, what proportion survives? We call this the *true normalized persistence ratio*, following the formulation of Lowe, Kim, Bondesan, and Hayakawa [8].

Quantum algorithms estimate Betti and persistent Betti numbers under access, state-preparation, density, and gap assumptions [7, 3]; complexity analyses show why the input and normalization conventions matter [11]. The ratio studied here sits between several nearby problems with different mathematical requirements. Exact clique homology can encode the multiplicity of a quantum satisfying space parsimoniously [1], yet the resulting Laplacian need not have an inverse-polynomial gap above all of its zero modes. Low-energy counting permits small positive eigenvalues and therefore does not determine an exact Betti number without a gap promise. Normalization by the number of simplices measures a different density. Finally, following one specified harmonic guide through a filtration [2] does not reveal the fraction of the complete initial homology that survives.

The technical obstruction is degeneracy. Existing clique-gadget analyses establish a spectral gap when a simulated logical Hamiltonian is positive definite [6]. A persistence reduction instead needs two logical Hamiltonians with known, generally nonzero kernels. It is insufficient to construct the desired zero modes. One must rule out every additional geometric zero mode and every positive eigenvalue below the promised threshold, including vectors supported largely outside the simulated register.

We address this obstruction through a finite-certificate all-chain theorem. The theorem assumes a fixed palette of clique gadgets. Each gadget supplies exact filling data, a zero-weight kernel decomposition, coercivity of a projected private differential, a local positive spectral floor, and a bounded interface with the common register. Unused register factors are padded by their harmonic spaces in every chain bidegree. These conditions are finite and can be checked exactly for a fixed palette. They imply, for every unit geometric chain x,

$$\operatorname{dist}(x, K_A)^2 \le C\left[t_A\lambda^2 + \frac{\langle x, \Delta_A x\rangle}{g_A\lambda^\kappa}\right], \qquad \kappa \le 26, \tag{1}$$

where K_A is the embedded kernel of the logical projector sum H_A, g_A is its smallest positive eigenvalue, and Δ_A is the full geometric Laplacian. The estimate concerns arbitrary geometric vectors; it is not a statement about distance to the geometric Laplacian's own kernel.

The form of Equation (1) matters quantitatively. A direct use of the earlier many-gadget master inequality would choose λ proportional to g_A, producing a high power of the logical gap in the final estimate. Here the structural error is $t_A\lambda^2$, while the energy term is divided by $g_A\lambda^\kappa$. We may choose λ from the palette size and target angle, independently of g_A. For locality at most six, a conservative choice gives

$$E_A = \Omega\left(\frac{\eta^{28} g_A}{t_A^{26}}\right).$$

The exponent is poor, but the scale is inverse polynomial and linear in the logical gap.

Exact filling then turns the analytic estimate into an exact topological statement. Let $V = Z_d(R)$ be the cycle space of the top-dimensional register and let W_A be the span of the logical projector ranges attached by the term set A. Independent gadget interiors give

$$B_d(X_A) \cap V = W_A.$$

This supplies an injection $V/W_A \hookrightarrow H_d(X_A)$. Equation (1) bounds the whole low-energy spectral subspace in the reverse direction, yielding

$$H_d(X_A) \cong V/W_A, \qquad \dim\ker\Delta_A = \dim\ker H_A, \qquad \operatorname{spec}(\Delta_A) \cap (0, E_A) = \varnothing.$$

For $A \subseteq B$, simplicial inclusion becomes the quotient epimorphism $V/W_A \twoheadrightarrow V/W_B$. The persistent rank is therefore exactly $\dim\ker H_B$. This quotient description avoids any need to choose harmonic representatives compatibly across the filtration.

Our complexity application uses an exact perfect-completeness circuit source. A three-bit mixed label transforms a G_2-verifier with acceptance probability p_x into the compressed operator

$$M_x = \operatorname{diag}(1, p_x, p_x, p_x, p_x, p_x, 0, 0).$$

In a YES instance $p_x = 1$, its perfect eigenspace has dimension six; in a NO instance $p_x \le 1/3$, that dimension is one. Combining the circuit history with the transfer theorem produces nested weighted clique complexes with

$$\beta_d(X_{\text{in}}) = 8, \qquad \operatorname{rank}[H_d(X_{\text{in}}) \to H_d(X_{\text{out}})] = \begin{cases} 6, & x \in L, \\ 1, & x \notin L. \end{cases}$$

The normalized values are 3/4 and 1/8. The hardness source is the exact gate-dependent class $\mathsf{BQP}_1^{G_2}$, with $G_2 = \{X, \mathsf{CX}, \mathsf{CCX}, H \otimes H\}$.

Finally, we use Hayakawa's common-copy blow-up [4] with identical labeled copy blocks at every filtration level. The symmetric sector is an isometric copy of the weighted complex, up to a common Laplacian scale, and the asymmetric sector has a separate positive floor. The common labeling makes the isometries commute with inclusion. Consequently, endpoint Betti numbers, persistent rank, the normalized ratio, and whole-kernel gaps transfer to unweighted clique filtrations. This step is a sourced corollary rather than a new unweighting mechanism.

Our contributions are:

1. a finite-certificate all-chain leakage theorem for degenerate logical kernels, with a gadget scale independent of the logical gap;
2. exact kernel multiplicity and a whole-positive-spectrum gap obtained from the all-chain estimate and exact filling;
3. a functorial quotient realization for nested projector term sets, which computes the induced homology rank without compatible harmonic representatives;
4. a fixed-eight, gate-dependent BQP_1-hardness theorem for true normalized persistence, together with a common-copy unweighted corollary.

The scope is intentionally narrow. Exact multiplicity, particular whole-kernel gaps, the normalized target, the individual few-term gadgets, and single-level unweighting all have close precedents. Our proof contribution is the corrected finite-certificate control of arbitrary geometric chains and its integration with a natural quotient filtration. We make no claim of ordinary BQP, unrestricted SDQC_1, arbitrary-gate BQP_1, or complex-phase hardness. We also do not claim that the inverse-polynomial gap is useful for near-term computation.

The rest of the paper defines the geometric and logical models, proves the transfer theorem, derives quotient naturality, instantiates the circuit and fixed palette, and transfers the construction to unweighted graphs. The appendices give the all-chain inequalities and the exact source calculations in expanded form.

2 Preliminaries and problem statement

All chain spaces are finite-dimensional over $\mathbb{C}$. A vertex-weighted simplicial complex has positive vertex weights and product simplex weights. In normalized simplex coordinates the boundary is a diagonal rescaling of ordinary incidence. We use

$$\Delta_{X,d} = \partial^*_{X,d}\partial_{X,d} + \partial_{X,d+1}\partial^*_{X,d+1}, \quad Z_d(X) = \ker \partial_{X,d}, \quad B_d(X) = \operatorname{ran} \partial_{X,d+1}.$$

Finite-dimensional Hodge theory identifies $H_d(X) = Z_d(X)/B_d(X)$ with $\ker \Delta_{X,d}$. For a positive semidefinite operator A, write $\gamma_+(A) = \min(\operatorname{spec}(A) \setminus \{0\})$, with the convention that a claimed lower bound on $\gamma_+(A)$ concerns the positive spectrum only. All certified boundary matrices and fillings below are rational. Rank, kernel dimension, image intersection, and filling validity are unchanged under scalar extension $\mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$, and the real positive-semidefinite inequalities complexify verbatim. We state homology over $\mathbb{C}$ while certifying the finite palette over $\mathbb{Q}$.

For $X_1 \subseteq X_2$, define

$$\beta_d(X_1 \to X_2) = \operatorname{rank}[H_d(X_1) \to H_d(X_2)], \qquad q_d(X_1, X_2) = \frac{\beta_d(X_1 \to X_2)}{\beta_d(X_1)}. \tag{2}$$

▶ **Definition 2.1** (Gapped true normalized persistence). *An instance of the weighted problem consists of explicit simple graphs $G_1 \subseteq G_2$ on one labeled vertex set, one common positive dyadic weight function $w : V \to \mathbb{Q}_{>0}$, a degree $d \geq 1$, positive rational lower bounds γ_1, γ_2, and a positive rational accuracy ε, all encoded in binary. Put $X_i = \mathrm{Cl}(G_i)$, using the weighted-boundary convention above. The promises are*

$$\beta_d(X_1) > 0, \qquad \operatorname{spec}(\Delta_{X_i,d}) \subseteq \{0\} \cup [\gamma_i, \infty) \quad (i = 1, 2).$$

The task is to output $\widetilde{q}$ such that $\Pr[|\widetilde{q} - q_d(X_1, X_2)| \leq \varepsilon] \geq 2/3$. Input size is the total bit length of the graph encodings, weights, d, the two gap bounds, and ε. The unweighted problem is the restriction $w \equiv 1$. Its fixed decision version distinguishes $q_d \geq 3/4$ from $q_d \leq 1/8$.

The denominator is the initial Betti number, not the number of simplices, and the statistic counts exact zero modes rather than eigenvalues below a cutoff.

The construction starts from a top-dimensional register complex R, so $C_{d+1}(R) = 0$, and sets $V = Z_d(R)$. The register is a join of normalized bowtie cycles, giving an explicit isometric computational-basis embedding. Each constraint is rank one on a constant-size active register. After joining with unaffected register factors it becomes an orthogonal projector Π_j on V, generally of higher rank. For a term set A, put

$$H_A = \sum_{j \in A} \Pi_j, \qquad K_A = \ker H_A, \qquad W_A = \sum_{j \in A} \operatorname{ran} \Pi_j = K_A^\perp. \tag{3}$$

We use a certified logical gap

$$H_A \succeq g_A(P_V - P_{K_A}), \qquad 0 < g_A \leq 1. \tag{4}$$

Attaching the weighted clique gadget for every $j \in A$ gives X_A. Register vertices have weight one, private vertices have weight λ, and distinct gadgets have independent private interiors. Δ_A denotes the full geometric Laplacian on all degree-d chains. Thus H_A and Δ_A act on different spaces.

For the circuit application, J inserts the fixed clean string around a maximally mixed logical input, U is an exact circuit, and

$$M = J^* U^* P_{\mathrm{acc}} U J, \qquad S = \ker(I - M).$$

The initial history kernel will have the full input dimension, while the final history kernel will be isomorphic to the perfect accepting space S.

3 Finite-certificate transfer to the complete geometric kernel

We isolate finite local conditions under which a clique-gadget construction controls every chain in the full geometric complex. The statement does not use convergence of a spectral projector whose rank grows with the simulated circuit.

3.1 The certificate model

Consider one active gadget before padding. Its weight-one register is R_a, its target cycle space is $V_a = Z_{d_a}(R_a)$, and every private vertex has weight λ. In normalized chain coordinates put

$$D_a(\lambda) = \begin{bmatrix} \partial_{d_a}(\lambda) \\ \partial_{d_a+1}(\lambda)^* \end{bmatrix} = D_{a,0} + \lambda D_{a,1}.$$

The value at $\lambda = 0$ is a matrix limit, not a singular change of basis.

▶ **Assumption 3.1** (Finite local certificate). *For every member a of a fixed finite palette there are an orthogonal subspace Q_a contained in the span of private simplex coordinates, an output projector R_a^{out}, and a rank-one projector π_a on V_a such that:*

(i) $B_{d_a}(Y_a) \cap V_a = \operatorname{ran} \pi_a$ *for every* $\lambda > 0$*;*

(ii) $\ker D_{a,0} = V_a \oplus Q_a$*, orthogonally;*

(iii) *for* $T_a(\lambda) = R_a^{\mathrm{out}} D_a(\lambda)$*,*

$$T_a(\lambda) = \lambda T_{a,1}, \qquad T_{a,1} V_a = 0, \qquad \|T_{a,1} q\| \geq b\|q\| \quad (q \in Q_a);$$

(iv) *every retained output coordinate contains a private vertex.*

The constant $b > 0$, the positive singular-value floor of $D_{a,0}$, and all matrix norms are uniform over the palette.

All data in Assumption 3.1 are finite integer or rational linear algebra. Exact fillings prove one inclusion in (i); matching boundary ranks and surviving register cosets, or the guard connecting-map argument, prove the reverse inclusion. Modular rank bounds with matching rational kernels certify (ii), and a nonzero Gram determinant certifies (iii). The explicit palette used later is recorded in Appendix C.

▶ **Lemma 3.2** (Weight scaling and harmonic padding). *Suppose $d_a = 2m_a - 1$ and $m_a \le m$. There is a palette constant $c > 0$ such that*

$$\gamma_+(\Delta_{Y_a,d_a}(\lambda)) \ge c\lambda^\kappa, \qquad \kappa := 4m + 2. \tag{5}$$

If Y_a is joined with unaffected register factors whose reduced homology is concentrated in top degree, then in the global target degree the zero-weight kernel is

$$V \oplus (Q_a \otimes \mathcal{H}_{\mathrm{out}}),$$

its orthogonal complement has a uniform positive floor, and the padded exact boundary intersection is the range of $\Pi_a = \pi_a \otimes I_{\mathcal{H}_{\mathrm{out}}}$ on V.

Proof. See Appendix A.1. ◀

The harmonic restriction in this lemma is essential. If the padded bulk contained every coordinate with a local private vertex, deleting a weight-one outside vertex could remain in that bulk with coefficient one. We instead keep local private coordinates tensored with outside harmonics and place every nonharmonic outside factor in the separately gapped sector.

3.2 Global theorem

Attach t_A padded gadgets to the common top-dimensional register R. Their private interiors are disjoint. Let L_j map the private input block of gadget j to the shared register output and assume the finite interface bound

$$L_j L_j^* \preceq N_j \lambda^2 I, \qquad N_j \le N_*. \tag{6}$$

The input blocks are orthogonal, while the outputs may overlap. Hence, for $L = [L_1 \ \cdots \ L_{t_A}]$,

$$LL^* = \sum_j L_j L_j^* \preceq N_* t_A \lambda^2 I.$$

▶ **Theorem 3.3** (Finite-certificate degenerate-kernel transfer). *Fix a palette satisfying Assumption 3.1 and Lemma 3.2 and Equation (6). Use outside-harmonic padding in every bidegree and independent private interiors. Let $\mathcal{A}$ be a finite nested family of term sets, let $t_{\max} = \max\{1, \max_{A\in\mathcal{A}} |A|\}$, and suppose*

$$H_A = \sum_{j\in A} \Pi_j \succeq g_A(P_V - P_{K_A}), \qquad K_A = \ker H_A, \qquad 0 < g_A \le 1$$

for every $A \in \mathcal{A}$. There are palette constants $c_0, C > 0$ such that, whenever $0 < \lambda \le c_0/t_{\max}$, every unit $x \in C_d(X_A)$ satisfies

$$\boxed{\|(I - P_{K_A})x\|^2 \le C\left[t_A\lambda^2 + \frac{\langle x, \Delta_A x\rangle}{g_A\lambda^\kappa}\right].} \tag{7}$$

Here P_{K_A} is the projector onto the embedded logical kernel, extended by zero outside the register.

For $0 < \eta < 1$*, choose additionally*

$$\lambda \le c_1 \min\{t_{\max}^{-1}, \eta/\sqrt{t_{\max}}\}, \qquad E_A \le c_2 \eta^2 g_A \lambda^\kappa. \tag{8}$$

Then

$$\dim\ker \Delta_A = \dim K_A, \qquad \operatorname{spec}(\Delta_A) \cap (0, E_A) = \varnothing. \tag{9}$$

This includes $K_A = 0$*.*

3.2.0.1 Proof architecture.

The full calculation is in Appendix A. The shared-output identity above and the private projected pairs first give the geometric leakage estimate

$$\|(I - P_V)x\|^2 \le C\left[t_A\lambda^2 + (t_A + \lambda^{-2})\langle x, \Delta_A x\rangle\right]. \tag{10}$$

Exact filling and Lemma 3.2 separately give

$$\Delta_A^\uparrow \succeq c\lambda^\kappa H_A^{\mathrm{emb}}. \tag{11}$$

Combining these inequalities with the logical gap proves Equation (7). The bound applies to every vector in a low-energy spectral subspace. Projection onto K_A is therefore injective on that subspace, so its dimension is at most $\dim K_A$. Independent exact fillings supply $\dim K_A$ zero modes through V/W_A; equality and the whole-positive-spectrum gap follow. No relative-acyclicity assumption is used.

For the explicit history palette $m \le 6$, so $\kappa = 26$. Taking a dyadic $\lambda = \Theta(\eta/t_{\max})$ gives the conservative scale

$$E_A = \Omega\left(\frac{\eta^{28} g_A}{t_{\max}^{26}}\right). \tag{12}$$

The exponent is not claimed to be optimal.

4 Filtered quotient naturality

The transfer theorem determines the dimension of every endpoint kernel. Exact filling determines the maps between those kernels at the homology level. This separation is useful: spectral concentration proves that there are no extra classes, while the quotient model computes persistence without comparing harmonic representatives.

▶ **Lemma 4.1** (Independent filling)**.** *For every term set* A*,*

$$B_d(X_A) \cap V = W_A = \sum_{j\in A} \operatorname{ran}\Pi_j. \tag{13}$$

Proof. Every vector in $\operatorname{ran}\Pi_j$ has a filling in gadget j, so $W_A \subseteq B_d(X_A) \cap V$. Conversely, write a degree-$(d+1)$ chain as a sum of its gadget components. If its total boundary lies in V, then its boundary has zero coordinate on every private d-simplex. Private supports of distinct gadgets are disjoint, so this vanishing holds separately for each gadget component. The remaining register boundary of component j lies in $B_d(Y_j) \cap V = \operatorname{ran}\Pi_j$. Summing over j proves the reverse inclusion. ◀

No relative-acyclicity hypothesis appears in Lemma 4.1. Extra geometric homology outside V is excluded later by the all-chain concentration theorem, not by the boundary decomposition alone.

▶ **Theorem 4.2** (Natural quotient realization). *Under the hypotheses and parameter choice of Theorem 3.3, the map*

$$q_A : V/W_A \longrightarrow H_d(X_A), \qquad q_A(v + W_A) = [v], \tag{14}$$

is an isomorphism. For $A \subseteq B$, these isomorphisms make the diagram

$$\begin{tikzcd} V/W_A \arrow[r, "\rho_{AB}"] \arrow[d, "q_A"', "\cong"] & V/W_B \arrow[d, "\cong"', "q_B"] \\ H_d(X_A) \arrow[r, "(\iota_{AB})_*"] & H_d(X_B) \end{tikzcd}$$

commute, where ρ_{AB} is the natural quotient map. Consequently,

$$\operatorname{rank}(\iota_{AB})_* = \dim(V/W_B) = \dim K_B. \tag{15}$$

Proof. See Appendix A.6. ◀

For a chain $A_0 \subseteq A_1 \subseteq \cdots \subseteq A_s$, every arrow in degree d is an epimorphism. The constructed barcode therefore has deaths but no new births in this degree. More generally, for any $i \leq j$,

$$\operatorname{rank}\big[H_d(X_{A_i}) \to H_d(X_{A_j})\big] = \dim K_{A_j}.$$

Harmonic representatives may rotate between filtration levels; the exact statement concerns the algebraic homology maps.

Taking A_{in} to be the clock, propagation, and clean-input terms of a history Hamiltonian, and taking A_{out} to add the final rejection term, gives

$$q_d(X_{\text{in}}, X_{\text{out}}) = \frac{\dim K_{\text{out}}}{\dim K_{\text{in}}}. \tag{16}$$

This identity is the bridge from exact logical perfect-space fractions to true normalized persistence.

5 Exact histories, a fixed clique palette, and hardness

Let

$$G_2 = \{X, \mathsf{CX}, \mathsf{CCX}, H \otimes H\}.$$

Following Rudolph's gate-dependent definition [10, Definition 2.2], $\mathsf{BQP}_1^{G_2}$ consists of promise problems decided by a polynomial-time uniform family of exact G_2 circuits on an all-zero input, with one computational-basis output bit, perfect completeness, and inverse-polynomial rejection probability in the no case. Rudolph's exact error reduction [10, Theorem 2.3] preserves perfect completeness and stays in any gate set containing G_2; we apply it first so that soundness is at most $1/3$. The superscript is essential: no approximate compilation or gate-independent identification is used below.

5.1 Rational histories and the certified palette

Write $K = \sqrt{2}H$. Replace each $H \otimes H$ transition by a gain edge K on one work bit followed by a loss edge $K^{-1} = H/\sqrt{2}$ on the other. Their composite is exactly $H \otimes H$, and each edge is enforced by two orthonormal rational three-term vectors. The vectors and their direct calculation appear in Appendix B.1. This construction is due to Rudolph [10].

The propagation constraints for $X, \mathsf{CX}, \mathsf{CCX}$ decompose, on their constant-size active support, into two-time vectors

$$(|0, z\rangle - |1, Uz\rangle)/\sqrt{2}.$$

Clock, input, and output penalties are basis projectors. Unaffected work qubits are identities and are implemented by the harmonic padding of Lemma 3.2; they are not expanded into exponentially many rank-one terms.

▶ **Lemma 5.1** (Certified fixed palette). *Every local constraint in the split G_2 history is implemented by a member of one fixed family of weighted clique gadgets of total support locality at most six. Every member satisfies Assumption 3.1 and Equation* (6), *with uniform constants, and the graph construction is polynomial in the circuit size.*

Proof. See Appendix C. It records the finite integer certificates, exact relabelings, selected-cycle guard closure, and polynomial-size bookkeeping. ◀

5.2 Complete history kernels

Let L be the number of legal clock positions after splitting the Hadamard transitions. Put $s_t = \sqrt{2}$ immediately after a gain edge and $s_t = 1$ otherwise. If V_t is the unitary prefix after removing these scalars, then

$$\mathcal{V}v = Z^{-1/2} \sum_{t=0}^{L-1} s_t |t\rangle V_t v, \qquad Z = \sum_t s_t^2, \qquad L \le Z \le 2L. \tag{17}$$

Every gain edge is immediately followed by its matching loss edge, with no gate between them, so $s_0 = s_{L-1} = 1$.

▶ **Proposition 5.2** (History kernels and gaps). *Let C be the valid clean-input space, $D = \dim C > 0$, and let $M = J^* U^* P_{\mathrm{acc}} U J$ be the compressed acceptance operator. Put $S = \ker(I - M)$. The initial history Hamiltonian has complete kernel $\mathcal{V}C$, while adding the final rejection projector gives complete kernel $\mathcal{V}S$. If $M|_{S^\perp} \preceq I/3$, then*

$$g_{\mathrm{in}} \ge \frac{1}{8L^2}, \qquad g_{\mathrm{out}} \ge \frac{1}{3L(8L^2+1)} \ge \frac{1}{27L^3}. \tag{18}$$

The second statement includes $S = 0$.

The proof in Appendix B uses a weighted path Poincaré inequality. The output compression is exactly

$$\mathcal{V}^* H_{\mathrm{rej}} \mathcal{V} = (I - M)/Z, \tag{19}$$

and a two-positive-operator estimate turns this angle penalty into the second gap in Equation (18). Combining Theorems 3.3 and 4.2, Lemma 5.1, , and Proposition 5.2 yields weighted clique complexes $X_{\text{in}} \subseteq X_{\text{out}}$ with

$$\beta_d(X_{\text{in}}) = D, \qquad \beta_d(X_{\text{out}}) = \beta_d(X_{\text{in}} \to X_{\text{out}}) = \dim S, \tag{20}$$

and inverse-polynomial gaps above both complete kernels.

5.3 A fixed-eight source

Given a $\mathsf{BQP}_1^{G_2}$ verifier Q_x with acceptance probability p_x, add a maximally mixed label $z \in \{0,1\}^3$, run Q_x unconditionally, and accept according to

$$[z = 000] \ \vee \ \big(q \wedge [z \in \{001, 010, 011, 100, 101\}]\big),$$

where q is the output of Q_x. The reversible predicate uses only $X, \mathsf{CX}, \mathsf{CCX}$, preserves z, and uncomputes its scratch. Its compressed acceptance operator on the label is exactly

$$M_x = \operatorname{diag}(1, p_x, p_x, p_x, p_x, p_x, 0, 0). \tag{21}$$

Thus the perfect eigenspace has dimension six when $p_x = 1$ and dimension one when $p_x \le 1/3$; every off-perfect eigenvalue in the latter case is at most $1/3$.

▶ **Theorem 5.3** (Fixed-eight hardness). *There is a deterministic polynomial-time many-one reduction from* $\mathsf{BQP}_1^{G_2}$ *to the decision version of weighted gapped true normalized persistence. Every output satisfies*

$$\beta_d(X_{\text{in}}) = 8, \qquad \beta_d(X_{\text{in}} \to X_{\text{out}}) = \begin{cases} 6, & x \in L, \\ 1, & x \notin L, \end{cases}$$

and both endpoint degree-d Laplacians have inverse-polynomial gaps above their complete kernels. Hence the normalized values are $3/4$ *and* $1/8$*. Additive error* $1/24$*, with decision threshold* $7/16$*, distinguishes the two cases with success probability at least* $2/3$.

Proof. Apply Equation (20) to Equation (21). The two ratios differ by $5/8$, and $1/24 < 5/16$. The logical gaps in Equation (18) are explicit rationals. The fixed palette supplies hard-coded positive rational lower bounds for the constants in Theorem 3.3. Fix $\eta_0 = 1/10$; from the polynomially bounded $t_{\max}$, choose the first integer b for which $\lambda = 2^{-b}$ satisfies both inequalities in Equation (8). For each endpoint output the largest dyadic rational E_i no greater than one half of $c_2 \eta_0^2 g_i \lambda^{26}$. Thus $b = O(\log t_{\max})$, and the binary encodings of λ and E_i have polynomial length. The weighted graph has $O(n + t_{\max})$ vertices and at most $O(n^2 + t_{\max} n)$ explicitly generated edges, because the register and every local interior are explicit and locality is constant. Hence the map outputs the two graphs, their common weight function, d, E_1, E_2, and the fixed decision threshold in deterministic polynomial time. ◀

For every fixed finite gate family covered by Rudolph's Theorem 3.4 [10], the exact simulation into G_2 may be composed with this reduction. This is a separate gate-family-relative statement, not gate-independent BQP_1-hardness.

The exact trace-versus-perfect-space obstruction and the mixed-spectator extension to individual real Hadamards are recorded in Appendix B.5. Neither is needed for Theorem 5.3.

6 A natural unweighted corollary

The weighted construction uses only weights 1 and a dyadic inverse-polynomial λ. Hayakawa's common-copy construction replaces each vertex v by a clique of f_v labeled copies and each base edge by all cross-block edges [4]. Choose one common $F = \lambda^{-2}$ and

$$f_v = Fw(v)^2.$$

Register vertices then have F copies and private vertices have one.

In degree k, let $U_{i,k}$ map a normalized base simplex to the normalized average of its one-copy-per-block lifts. Lemma 4.2 and Theorem 4.3 of Hayakawa [4] give the symmetric identity and asymmetric gap over real chains; scalar extension gives the same statements over $\mathbb{C}$:

$$\widehat{\Delta}_{i,k} U_{i,k} = F U_{i,k} \Delta_{i,k}, \qquad \widehat{\Delta}_{i,k}|_{\mathrm{ran}(U_{i,k})^\perp} \succeq \min_v f_v \, I. \tag{22}$$

Use the same labeled copy block for every shared vertex at both filtration levels. With consistent orientations, simplicial inclusion then satisfies

$$\widehat{J}_{ij} U_{i,d} = U_{j,d} J_{ij}. \tag{23}$$

There is no extra normalization factor because an inclusion preserves the simplex degree, weights, labels, and copy set.

▶ **Corollary 6.1** (Unweighted true normalized persistence). *The reduction in Theorem 5.3 maps in polynomial time to nested unweighted clique complexes with the same endpoint Betti numbers, persistent rank, and normalized values $3/4$ and $1/8$. Both unweighted endpoint Laplacians retain inverse-polynomial gaps above their complete kernels.*

Proof. The asymmetric bound in Equation (22) excludes asymmetric harmonic vectors, so

$$\ker \widehat{\Delta}_{i,d} = U_{i,d} \ker \Delta_{i,d}.$$

Together with Equation (23), this makes every inclusion-induced map on harmonic spaces unitarily equivalent to its weighted counterpart. Endpoint nullities and induced ranks are therefore preserved. If E_i lower-bounds the positive weighted spectrum, then

$$\gamma_+(\widehat{\Delta}_{i,d}) \geq \min\{FE_i, \min_v f_v\} = \min\{FE_i, 1\}.$$

By Equation (12), $FE_i = \Omega(\eta^{26} g_i / t_{\max}^{24})$. Since F is polynomial, so is the blown-up graph. More explicitly, if the weighted graph has $O(n + t_{\max})$ vertices and $O(n^2 + t_{\max} n)$ edges, a safe explicit bound after blow-up is $O(F(n + t_{\max}))$ vertices and $O(F^2(n^2 + t_{\max} n))$ edges; labels and edges are generated by nested loops in polynomial time. Future private vertices may be present initially as isolated vertices; the target degree is positive, so they add no target-degree chain. ◀

The operator decomposition is Hayakawa's; the filtration-specific step is the commuting identity Equation (23). We do not count the unweighting mechanism as a separate contribution.

7 Relation to prior work

Crichigno and Kohler give a parsimonious reduction in which quantum satisfying states correspond to clique-homology classes one-for-one [1]; exact multiplicity alone is therefore prior. King and Kohler develop the weighted clique gadgets and many-gadget spectral analysis behind gapped clique homology [6]. Once a uniform arbitrary-chain estimate is available, substituting a gap above a degenerate logical kernel and applying the dimension argument of Theorem 3.3 are short consequences.

Our independent estimate changes the padded step. The coordinate-bulk argument in arXiv version 2 [5] can retain a term obtained by deleting a weight-one outside vertex, so its coefficient need not be $O(\lambda)$. We project the fixed private pair onto outside harmonics in every bidegree and give all nonharmonic outside modes a separate positive floor. Direct finite matrices also avoid turning basiswise perturbation into a dimension-dependent projector bound. This comparison is only to the cited arXiv version; the theorem here is proved independently and makes no claim about the presentation in the published SIAM article.

Gyurik et al. prove gate-dependent hardness for survival of one specified harmonic guide and state a complete-final-kernel gap in their construction [2]. Our statistic instead asks for the fraction of the full initial homology that survives, and our all-chain estimate is used at both endpoints. Lowe et al. define this initial-Betti-normalized quantity and formulate the exact-kernel, gap, and compatibility requirements for clique filtrations [8]. The quotient maps V/W_A meet the required rank interface for our restricted family without choosing compatible harmonic representatives. Persistent Laplacians provide the general spectral formulation of induced homology ranks [9]; we use ordinary endpoint Laplacians for the gap promise and the quotient model for the map.

Rudolph supplies the exact gain/loss vectors, real-gate simulations, and base clique constructions used in the certified palette [10]. Hayakawa's symmetric/asymmetric decomposition supplies the unweighting mechanism [4]. Our filtration-level addition is the common-label commuting identity Equation (23); it is a direct corollary, not a separate unweighting theorem.

8 Limitations and conclusion

The construction has three substantive limits. Its exact palette is finite and real; arbitrary integer states and complex phases are outside the theorem. Its hardness source is the gate-dependent class $\mathsf{BQP}_1^{G_2}$, rather than ordinary BQP, unrestricted SDQC_1, or a gate-independent perfect-completeness class. Finally, the clean source has $\beta_d(X_{\mathrm{in}}) = 8$. Ignored mixed qubits can replicate this denominator exactly, but an intrinsically growing family of distinct initial classes would require another exact source.

The gap is also a complexity promise rather than a practical estimate. At locality six it scales conservatively as $\Omega(\eta^{28} g/t^{26})$, with $g = \Omega(L^{-3})$ for the history construction. The common-copy blow-up is polynomial but may be large.

Within these limits, finite local certificates control every geometric chain; exact fillings supply the intended zero modes; the all-chain estimate excludes extra zero and low-energy modes; and natural quotients compute the induced homology rank. The reusable result is the transfer criterion itself: a new clique reduction can obtain gapped exact persistence by certifying local zero-weight kernels, private coercivity, harmonic padding, filling, and interfaces, without a growing-dimensional perturbation argument.

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A Proof of the all-chain transfer theorem

This appendix proves Theorem 3.3 from the finite local hypotheses. All constants below depend only on the fixed palette.

A.1 Local scaling and harmonic padding

Let W_k be the diagonal matrix of simplex weights. In normalized coordinates,

$$\partial_{w,k} = W_{k-1}^{-1}\partial_{1,k}W_k.$$

For $0 < \lambda \leq 1$, invertible left and right scaling at the fixed rank of $\partial_{1,k}$ gives

$$\sigma^+_{\min}(\partial_{w,k}) \geq \lambda^{k+1}\sigma^+_{\min}(\partial_{1,k}).$$

Indeed, put $K = \ker\partial_{1,k}$. For every unit $x \perp W_k^{-1}K$,

$$\operatorname{dist}(W_k x, K) \geq \sigma_{\min}(W_k).$$

Apply the least nonzero singular value of the unweighted boundary. The other diagonal factor satisfies

$$\sigma_{\min}(W_{k-1}^{-1}) \geq 1.$$

The Hodge decomposition applies these estimates separately to the down- and up-exact sectors and gives

$$\gamma_+(\Delta_{Y_a,d_a}(\lambda)) \geq c_a\lambda^{2d_a+4}.$$

A finite palette and $d_a = 2m_a - 1$, $m_a \leq m$, give the common bound Equation (5).

For padding, use augmented reduced chain complexes, including degree -1, in every join bidegree:

$$\widetilde{C}_d(Y_a * R_{\mathrm{out}}) = \bigoplus_{r+s=d-1}\widetilde{C}_r(Y_a)\otimes\widetilde{C}_s(R_{\mathrm{out}}).$$

On each summand the join Laplacian is the tensor sum

$$\Delta_{Y_a * R_{\mathrm{out}}} = \Delta_{Y_a}\otimes I + I\otimes\Delta_{R_{\mathrm{out}}}.$$

The outside register has reduced homology only in its top degree and a constant positive gap on every other mode. At the global target degree, therefore, the only outside-harmonic bidegree is the active target degree tensored with $\mathcal{H}_{\mathrm{out}}$. This proves the stated zero-weight kernel and puts every other bidegree above a uniform floor. Tensor the private output projector with $P_{\mathcal{H}_{\mathrm{out}}}$; both outside differentials vanish after this projection, preserving the private-pair bounds.

Finally, the relative join formula identifies the padded connecting map with the active connecting map tensored with $\mathcal{H}_{\mathrm{out}}$. Since all register weights are one, the positive-weight diagonal gauge fixes the register coordinates. Thus the padded boundary intersection is $\operatorname{ran}(\pi_a \otimes I_{\mathcal{H}_{\mathrm{out}}})$, completing the proof of Lemma 3.2.

A.2 Global decomposition and interface energy

Write the target chain space as the orthogonal coordinate sum

$$C_d(X_A) = C_d(R) \oplus \bigoplus_{j\in A} C_{d,j}^{\mathrm{priv}}.$$

For a unit chain x, write

$$x = \omega_0 + \sum_j \omega_j, \qquad x_j = \omega_0 + \omega_j, \qquad U = \sum_j \|\omega_j\|^2, \qquad e = \langle x, \Delta_A x\rangle.$$

Let $a = \partial_R \omega_0$, $y_j = L_j\omega_j$, $y = \sum_j y_j$, and $r = a + y$. The vector r is the shared-register component of the global down-boundary. Since $a + y_j = r - (y - y_j)$,

$$\sum_j \|a + y_j\|^2 \le 2t_A\|r\|^2 + 2\sum_j \|y - y_j\|^2.$$

The elementary identity

$$\sum_j \|y - y_j\|^2 = (t_A - 2)\|y\|^2 + \sum_j \|y_j\|^2$$

also holds for $t_A = 1$. Put

$$\rho = \lambda^{-2}\|[L_1 \ \cdots \ L_{t_A}]\|^2 \le N_* t_A, \qquad \chi = (t_A - 2)_+\rho + N_* \le C t_A^2.$$

Then Equation (6) gives

$$\sum_j \|y - y_j\|^2 \le \chi\lambda^2 U.$$

All nonregister down-boundary coordinates and all up-boundary coordinates are private and partition by gadget. We obtain

$$\sum_j \|D_j(\lambda)x_j\|^2 \le 2t_A e + 2\chi\lambda^2 U. \tag{24}$$

A.3 Zero-weight and private-sector coercivity

Let $A_{j,0} = I - P_V - Q_j$ in the padded local chain space. The positive singular-value floor of $D_{j,0}$ on $\operatorname{ran} A_{j,0}$, together with $D_j(\lambda) = D_{j,0} + \lambda D_{j,1}$, gives

$$\begin{aligned} S_A := \sum_j \|A_{j,0}x_j\|^2 &\le C\sum_j \|D_{j,0}x_j\|^2 \\ &\le C\left(t_A e + \chi\lambda^2 U + \lambda^2\sum_j \|x_j\|^2\right) \\ &\le C\left(t_A e + \chi\lambda^2 U + t_A\lambda^2\right). \end{aligned} \tag{25}$$

The final inequality uses $\sum_j \|x_j\|^2 = t_A\|\omega_0\|^2 + U \le t_A$.

Because $T_j P_V = 0$,

$$T_j Q_j x_j = T_j x_j - T_j A_{j,0} x_j.$$

The lower bound on T_j restricted to Q_j, the estimate $\|T_j\| \le C\lambda$, and orthogonality of the retained private output blocks imply

$$S_Q := \sum_j \|Q_j x_j\|^2 \le C\left(\frac{e}{\lambda^2} + S_A\right). \tag{26}$$

The orthogonality here concerns the retained T_j outputs. The shared L_j outputs were handled by their horizontal operator norm and need not be orthogonal.

Let $N = \|(I - P_V)x\|^2$. Summing the orthogonal decompositions $P_V \oplus Q_j \oplus \operatorname{ran} A_{j,0}$ gives

$$N \le U + t_A \|(I - P_V)\omega_0\|^2 = S_A + S_Q.$$

Since $U \le N$, substituting Equations (25) and (26) and using $\chi \le C t_A^2$ yields

$$N \le C\left[t_A \lambda^2 + (t_A + \lambda^{-2})e\right] + C t_A^2 \lambda^2 N.$$

Choosing $\lambda \le c_0/t_{\max}$ absorbs the last term and proves Equation (10).

A.4 Logical coercivity

For each gadget, exact filling gives $\operatorname{ran} \Pi_j \subseteq B_d(Y_j)$. The up-Laplacian has kernel $B_d(Y_j)^\perp$, and Equation (5) lower-bounds its positive spectrum. Therefore, after the common embedding into X_A,

$$\Delta_j^\uparrow \succeq c\lambda^\kappa \Pi_j.$$

The top-dimensional register has no up-columns, and every degree-$(d+1)$ simplex with a private vertex belongs to exactly one gadget. Hence the global up-Laplacian is the exact sum of the embedded local up-Laplacians, proving Equation (11).

Since $K_A \subseteq V$, the squared distance decomposes as

$$\|(I - P_{K_A})x\|^2 = N + \langle x, (P_V - P_{K_A})x \rangle.$$

Equations (4) and (11) give

$$\langle x, (P_V - P_{K_A})x \rangle \le \frac{e}{c g_A \lambda^\kappa}.$$

Together with Equation (10),

$$\|(I - P_{K_A})x\|^2 \le C\left[t_A \lambda^2 + (t_A + \lambda^{-2})e + \frac{e}{g_A \lambda^\kappa}\right].$$

For $\lambda \le c_0/t_{\max}$, $g_A \le 1$, and $\kappa \ge 2$, the last energy term dominates the middle energy terms up to a palette constant. This proves Equation (7).

A.5 Exact multiplicity and the whole-positive-spectrum gap

Let $\mathcal{L}_A(E_A)$ be the span of all eigenvectors of Δ_A with eigenvalue below E_A. Every unit $x \in \mathcal{L}_A(E_A)$ obeys $e < E_A$. Choosing the constants in Equation (8) makes the right-hand side of Equation (7) strictly less than $\eta^2 < 1$. Therefore $P_{K_A} x \neq 0$ for every nonzero $x \in \mathcal{L}_A(E_A)$, so $P_{K_A}|_{\mathcal{L}_A(E_A)}$ is injective and

$$\dim \mathcal{L}_A(E_A) \le \dim K_A. \tag{27}$$

Every $v \in V$ remains a cycle after attaching the gadgets. By Lemma 4.1, the map $V/W_A \to H_d(X_A)$ is injective. Its domain has dimension $\dim K_A$, so Hodge theory supplies at least $\dim K_A$ zero eigenvectors of Δ_A. These zero vectors belong to $\mathcal{L}_A(E_A)$. Combining this lower bound with Equation (27) forces equality throughout: the geometric kernel has dimension $\dim K_A$, and no positive eigenvalue lies below E_A.

If $K_A = 0$, any unit vector in $\mathcal{L}_A(E_A)$ would have distance one from K_A, contradicting the strict $\eta < 1$ bound. Thus the same argument proves $\Delta_A \succeq E_A I$ in the zero-kernel case. No additional global relative-acyclicity assumption is required.

A.6 Proof of quotient naturality

Equation (13) makes q_A well-defined and injective. Its domain has dimension

$$\dim(V/W_A) = \dim K_A.$$

Hodge theory and Equation (9) give

$$\dim H_d(X_A) = \dim \ker \Delta_A = \dim K_A,$$

so q_A is also surjective. If $A \subseteq B$, both paths in the diagram of Theorem 4.2 send $v + W_A$ to the class of the same register cycle v in X_B. Thus the diagram commutes. The quotient map is onto because $W_A \subseteq W_B$, and its rank is $\dim(V/W_B) = \dim K_B$.

B History-gap and source calculations

B.1 Rational gain/loss split

On a clock bit c and work bit w, define

$$\begin{aligned} u_0 &= (|00\rangle + |01\rangle - |10\rangle)/\sqrt{3}, \quad u_1 = (|00\rangle - |01\rangle - |11\rangle)/\sqrt{3}, \\ d_0 &= (|00\rangle - |10\rangle - |11\rangle)/\sqrt{3}, \quad d_1 = (|01\rangle - |10\rangle + |11\rangle)/\sqrt{3}. \end{aligned}$$

The vectors in each pair are orthonormal. If consecutive history amplitudes are f_t, f_{t+1}, orthogonality to u_0, u_1 is equivalent to $f_{t+1} = \sqrt{2} H f_t$; orthogonality to d_0, d_1 is equivalent to $f_{t+1} = H f_t/\sqrt{2}$. Applying the first relation to one work bit and the second to the other gives $H \otimes H$.

B.2 Weighted path estimate

With the unary encoding $|t\rangle = |1^t 0^{T-t}\rangle$, the legal clock span and its orthogonal complement are invariant under the neighboring-clock-guarded propagation, input, and output projectors. The diagonal clock term

$$H_{\text{clock}} = \sum_{j=1}^{T-1} |01\rangle\langle 01|_{j,j+1}$$

has energy at least one on every illegal computational-basis clock string. By Hermiticity, the whole unary-clock Hilbert space is the orthogonal sum of the invariant legal and illegal sectors. It therefore suffices to prove the smaller inverse-polynomial bound on the legal sector. For a legal-clock state, write $f_t = s_t V_t \xi_t$, with the notation of Equation (17). Its norm and initial-history energy are

$$N(\xi) = \sum_t w_t \|\xi_t\|^2, \qquad E(\xi) = \sum_{t=1}^{L-1} c_t \|\xi_t - \xi_{t-1}\|^2 + \langle \xi_0, A_{\mathrm{in}} \xi_0 \rangle,$$

where $w_t = s_t^2 \in [1,2]$, $c_t \in \{1/2, 2/3\}$, and A_{in} is the projector onto $C^\perp$. Split every ξ_t into the clean subspace $C = \ker A_{\mathrm{in}}$ and its orthogonal complement.

For a state orthogonal to all valid histories, the clean component has weighted mean zero:

$$\sum_t w_t \xi_t^C = 0.$$

The weighted variance identity and path Cauchy–Schwarz inequality give

$$\sum_t w_t \|\xi_t^C\|^2 = \frac{1}{2\sum_t w_t} \sum_{i,j} w_i w_j \|\xi_i^C - \xi_j^C\|^2 \le 2L^2 E_C.$$

For the dirty component, set $a = \|\xi_0^{C^\perp}\|^2$ and $D_B = \sum_t \|\xi_t^{C^\perp} - \xi_{t-1}^{C^\perp}\|^2$. The bound

$$\|\xi_t^{C^\perp}\|^2 \le 2a + 2(L-1)D_B$$

implies

$$\sum_t w_t \|\xi_t^{C^\perp}\|^2 \le 8L^2 E_B.$$

Adding both sectors proves $N(\xi) \le 8L^2 E(\xi)$ on the orthogonal complement of the complete initial kernel, establishing the first bound in Equation (18).

B.3 Adding the output projector

We record the two-positive-operator estimate used for the second bound in Equation (18). Let P project onto $\ker A$, suppose $A \succeq g(I-P)$, and let R be an orthogonal projector. Put $\mathcal{S} = \ker A \cap \ker R$, and suppose $PRP \succeq \alpha P$ on $\ker A \ominus \mathcal{S}$. On every nontrivial canonical block for the pair P, R, for some $\rho \in [\alpha, 1]$,

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} \rho & \sqrt{\rho(1-\rho)} \\ \sqrt{\rho(1-\rho)} & 1-\rho \end{pmatrix}.$$

Thus $g(I-P)+R$ has trace $g+1$, determinant $g\rho$, and smallest eigenvalue

$$\frac{g+1-\sqrt{(g+1)^2-4g\rho}}{2}.$$

This quantity increases with ρ; the one-dimensional canonical blocks outside $\mathcal{S}$ obey the same lower bound. Since $A+R \succeq g(I-P)+R$ and both operators have kernel $\mathcal{S}$,

$$\gamma_+(A+R) \geq \frac{g+1-\sqrt{(g+1)^2-4g\alpha}}{2} \geq \frac{g\alpha}{g+1}. \tag{28}$$

For the history Hamiltonian, $g=1/(8L^2)$ and $\alpha=(1-r)/Z \geq 1/(3L)$ when $r \leq 1/3$. Substitution into Equation (28) gives

$$\gamma_+(H_{\text{out}}) \geq \frac{1}{3L(8L^2+1)}.$$

The kernel identity follows directly from positivity:

$$\ker(H_{\text{in}}+H_{\text{rej}}) = \ker H_{\text{in}} \cap \ker H_{\text{rej}} = \mathcal{V}\ker(I-M).$$

B.4 Exact eight-label compression

Let U_x denote the fixed-input verifier and q its output bit. The label z is never changed. After computing and uncomputing the constant-size acceptance predicate, compression through all clean work registers has no matrix element between distinct labels. On $z=000$, acceptance is the identity. On five labels it is the scalar $p_x = \|P_{\text{acc}}U_x|0\cdots0\rangle\|^2$, and on two labels it is zero. This proves Equation (21) even for a coherent vector across label values.

The normalized trace is

$$\frac{\operatorname{Tr} M_x}{8} = \frac{1+5p_x}{8}.$$

If $p_x=1$, this equals $3/4$ and coincides with the perfect-space fraction. If $p_x \leq 1/3$, it is at most $1/3$, the perfect-space fraction is exactly $1/8$, and every off-perfect eigenvalue is at most $1/3$. Thus the construction simultaneously supplies the trace promise, the perfect-space separation, and the off-perfect ceiling needed by the output-gap proof.

B.5 Source-interface boundary

The exact source used by Theorem 5.3 begins with a fixed-input $\mathsf{BQP}_1^{G_2}$ verifier. The three mixed label bits are added by the reduction and are the entire input space of Equation (21); the verifier's input and all predicate scratch remain clean. Consequently the initial history dimension is exactly eight. No trace-to-rank inference, unrestricted SDQC_1 identification, or approximate gate compilation enters the proof.

The finite gadget evidence needed to compile these circuit terms is given in Appendix C. The theorem uses those exact finite objects and does not infer their topology from floating-point spectra.

For an arbitrary acceptance operator with $M|_{S^\perp} \preceq rI$, put $f=\dim S/D$ and $p=\operatorname{Tr}(M)/D$. Spectral decomposition gives

$$f \leq p \leq r+(1-r)f.$$

Thus trace promises $p \geq a$ versus $p \leq b$ force a perfect-space fraction gap only when $\max\{0, (a-r)/(1-r)\} > b$. The eight-label source avoids this obstruction by fixing the perfect fractions directly.

An exact circuit over $\{X, \mathsf{CX}, \mathsf{CCX}, H\}$ admits the following real-gate extension. Add one maximally mixed, unmeasured spectator a and replace each H_j by $H_j \otimes H_a$. If there are h Hadamards, then

$$U' = U \otimes H_a^h, \qquad J' = J \otimes I_a, \qquad P'_{\mathrm{acc}} = P_{\mathrm{acc}} \otimes I_a.$$

For either parity of h,

$$(J')^*(U')^*P'_{\mathrm{acc}}U'J' = M \otimes I_a.$$

Both the input dimension and perfect eigenspace dimension double, while the fraction and off-perfect ceiling are unchanged. This calculation does not cover complex phases.

C Finite palette and guard certificates

This appendix identifies the finite objects used in Lemma 5.1. The underlying active graphs are four explicit constructors in the source artifact: `state_0m1`, `state_00m11`, `state_01m10`, and `state_00m10m11`. They accompany Rudolph's gain/loss construction [10]. The supplementary certificate archive records the complete graphs, oriented integer fillings, kernel bases, modular rank witnesses, and Gram determinants. No numerical eigenvalue threshold is used.

Active vector	degree	vertices	$\dim V$	$\dim Q$	certificate mod p
$\|0\rangle - \|1\rangle$	1	16	2	8	$\det G = 512$
$\|00\rangle - \|11\rangle$	3	33	4	112	$\det G = 655970$
$\|01\rangle - \|10\rangle$	3	33	4	112	signed relabeling
$\|00\rangle - \|10\rangle - \|11\rangle$	3	41	4	176	$\det G = 797443$
three remaining gain/loss vectors	3	41	4	176	signed relabelings

Table 1 Exact active-palette data. In each determinant row, $p = 1000003$ and $G = T_{a,1}^* T_{a,1}$ on Q_a; a nonzero residue proves injectivity over $\mathbb{Q}$.

For the one-qubit atom, exact boundary ranks are $15, 24$ and the Betti numbers are $(1, 1, 0)$; a denominator-one 24-term two-chain fills the target cycle. At zero weight the target differential pair has rank 30, and the two register cycles plus eight private vectors give its complete ten-dimensional kernel. For $|00\rangle - |11\rangle$, the clique counts in degrees zero through four are $(33, 241, 646, 712, 272)$, the boundary ranks are $(32, 209, 437, 272)$, and the Betti numbers are $(1, 0, 0, 3, 0)$. A denominator-one 248-term four-chain fills the target; rank 596 of the zero-weight pair matches the four register and 112 private kernel vectors. Swapping the second bowtie petals transports this certificate exactly to $|01\rangle - |10\rangle$.

For the three-term representative, the clique counts in degrees zero through five are

$$(41, 322, 914, 1082, 477, 30),$$

and the boundary ranks are $(40, 282, 632, 447, 30)$. Hence the target degree-three Betti number is three. A denominator-one 382-term four-chain fills $|00\rangle - |10\rangle - |11\rangle$, while three complementary register cycles remain independent modulo boundaries. The zero-weight pair has rank 902, matching its kernel $V \oplus Q$ of dimension $4 + 176$. Signed permutations of bowtie petals transport the graph, filling, kernel, and private pair to the other three gain/loss vectors.

C.1 Selected-cycle guards

We record the guard closure because it is part of the theorem interface, not an appeal to a general integer-state gadget. Let Y implement π_ϕ on register R, let B be a fresh bowtie with petals S_0, S_1, and choose $\beta \in \{0, 1\}$. Add all edges of B, join every vertex of R to every vertex of B, and join each private vertex of Y only to the selected petal S_β. The resulting flag complex is

$$Z = (Y * S_\beta) \cup (R * B), \qquad (Y * S_\beta) \cap (R * B) = R * S_\beta. \tag{29}$$

Its new register is $R' = R * B$. If γ_β is the normalized oriented cycle of the selected petal, then Z implements

$$\pi_\phi \otimes |\beta\rangle\langle\beta|, \qquad Q' = Q \otimes \mathbb{C}\gamma_\beta.$$

Indeed, reduced relative chains give

$$\widetilde{C}_*(Z, R') \cong \widetilde{C}_*(Y, R) \widehat{\otimes} \widetilde{C}_*(S_\beta).$$

The connecting map therefore tensors the old exact boundary intersection with $[\gamma_\beta]$, proving exact filling and no additional killed register class. At zero private weight the private differential is the corresponding tensor-sum differential. Since S_β has reduced homology only in degree one, the complete target kernel is $V' \oplus Q'$; every other bidegree pays a fixed positive guard gap. Finally, project the old private output pair onto the selected harmonic line. Both outside differentials vanish under this projection, so

$$T_Z(\lambda) = \lambda T_{Z,1}, \qquad T_{Z,1} V' = 0, \qquad \|T_{Z,1}(q \otimes \gamma_\beta)\| \geq b \|q \otimes \gamma_\beta\|.$$

Removing the unique old private vertex also gives $LL^* \preceq N_{\mathrm{priv}} \lambda^2 I$. Iterating this construction for the constant number of required basis guards preserves a fixed finite palette.

The basis projector itself is the special case $R \cup (v * S_\beta)$, where a private apex v cones the selected petal. Its private zero-weight kernel Q is zero, so the private-pair condition is vacuous. Together with Table 1, this proves Lemma 5.1.

C.2 Reproducibility boundary

The supplementary archive contains the exact witnesses in

- RUDOLPH_REPRESENTATIVE_BULK_CERTIFICATE.json,
- REMAINING_ACTIVE_ATOM_CERTIFICATES.json,
- ACTIVE_HADAMARD_ORBIT_CERTIFICATE.json, and
- SELECTED_CYCLE_GUARD_CHECKS.json.

Their checkers recompute the stated ranks, fillings, relabelings, and guard identities. These finite checks certify the displayed palette; the analytic implication from those certificates is the proof of Theorem 3.3.

AI Disclosure

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